



Computer Vision and Image Processing

Lecture 09

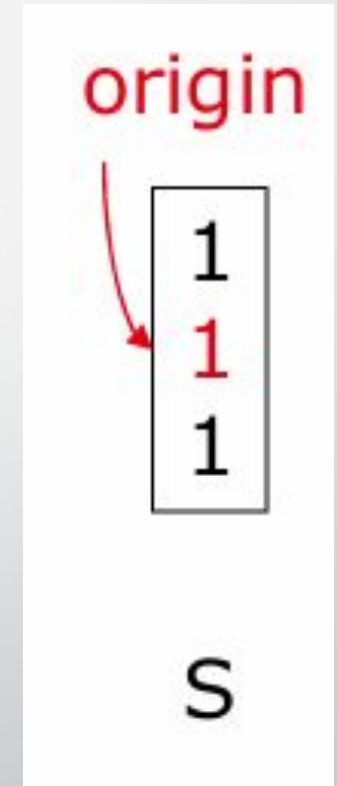
Morphology

- In digital image processing, morphology is a framework of non-linear operations used to modify, extract, or clean up the **shapes and structures** of objects within an image.
- It is most commonly applied to **binary images** (where pixels are either 0 or 1, black or white), but it can also be extended to **grayscale images**.

Morphology

The Structuring Element (SE)

- Morphological operations work by passing a small shape or template called a **Structuring Element** (or kernel) over the input image, much like a convolution matrix in linear filtering.
- The structuring element can be of any size and shape (e.g., a square, a circle, a cross, or a line). As it slides across the image, a logical comparison (like an AND or OR operation) is made between the structuring element and the underlying image pixels to determine the value of the output pixel.



Morphology

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

(a) BOX(3,5)

	1	1	1	
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
	1	1	1	

(b) DISK(5)

	1	1	1	
1				1
1				1
1				1
	1	1	1	

(c) RING(5)

1	1		
1	1		
1	1	1	1
1	1	1	1

(d)

1	1	1	1	1	1
1		1	1		1
1		1	1		1
1		1	1		1

(e)

1
1
1
1

(f)

Morphology

1. Fundamental Operations:

- a) Dilation
- b) Erosion

2. Composite Operations:

- a) Opening
- b) Closing
- c) Morphological Gradient

3. Top-Hat Transforms:

- a) White Top-Hat (Top-Hat)
- b) Black Top-Hat (Bottom-Hat)

Dilation

Dilation expands the boundaries of foreground objects, effectively making them **larger** and filling in small holes or gaps.

$$I = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$SE = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Dilation

$$I \oplus SE = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Erosion

Erosion shrinks the boundaries of foreground objects, effectively making them **smaller** and removing small objects or thin structures.

$$I = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$SE = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Erosion

$$I \ominus SE = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Opening

Opening consists of an **Erosion** followed by a **Dilation**.

$$I \circ SE = (I \ominus SE) \oplus SE$$

- **What it does:** It removes small bright objects (noise), breaks thin connections, and smooths sharp peaks from the *inside* of an object.
- **Key characteristic:** It preserves the overall shape and size of larger objects that can completely contain the structuring element.

Closing

Closing consists of a **Dilation** followed by an **Erosion**.

$$I \bullet SE = (I \oplus SE) \ominus SE$$

- **What it does:** It fills in small dark holes, fuses narrow gaps or breaks, and smooths contours from the *outside*.
- **Key characteristic:** It bridges gaps that are smaller than the size of the structuring element.

Morphological Gradient

The **Morphological Gradient** (or simply "Gradient") in image processing is a fundamental operation that calculates the **difference** between the **Dilation** and the **Erosion** of an image. It is often used to highlight the **boundaries** or **edges** of objects within the image.

$$G(I) = (I \oplus SE) - (I \ominus SE)$$

Morphological Gradient

Why It Highlights Edges

- **Dilation** expands the object, making the boundary thicker (the expansion is the region *outside* the original object).
- **Erosion** shrinks the object, making the boundary thinner (the shrinking is the region *inside* the original object).
- When you **subtract** the smaller, eroded image from the larger, dilated image, the only remaining pixels are the ones that were added by dilation and/or removed by erosion. These pixels form a **thickened outline** or **boundary** of the original object.

White Top-Hat Transform (WTH)

The White Top-Hat transform extracts **small, bright objects** or **details** from an image, especially when the background illumination is uneven.

$$\text{WTH}(I) = I - (I \circ SE)$$

Black Top-Hat Transform (BTH)

The Black Top-Hat transform (or Bottom-Hat) extracts **small, dark objects** or **details** from an image.

$$\text{BTH}(I) = (I \bullet SE) - I$$



The End